

Indian Startup Ecosystem Analytics

(Data Cleaning, Standardization & SQL-Based Business Analysis)

About the Dataset

Source: Kaggle

Dataset: [Indian Startups - Funding Data](#)

Objective

The goal of this project is to clean, standardize, and analyze startup funding data collected between 2018 and 2021. Since the data spans multiple years with inconsistent schemas and formatting, the first objective is to prepare a unified dataset suitable for analysis.

The project aims to answer questions such as:

- How has startup funding changed over the years?
- Which sectors received the highest investment?
- Which cities have the largest startup ecosystem?
- Which startups raised the most funding?
- How does funding vary across different investment stages?

The project includes data cleaning and preprocessing using Python (Pandas), followed by SQL-based exploratory analysis to uncover funding trends and business insights.

```
#importing necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import zipfile
import os

# Extracting the zip file
with zipfile.ZipFile('archive.zip', 'r') as zip_ref:
    zip_ref.extractall('startup_data')

#checking the files inside the zip file
print(os.listdir('startup_data'))

['startup_funding2020.csv', 'startup_funding2019.csv',
 'startup_funding2021.csv', 'startup_funding2018.csv']

#reading the csv files
df18 = pd.read_csv('startup_data/startup_funding2018.csv')
df19 = pd.read_csv('startup_data/startup_funding2019.csv')
df20 = pd.read_csv('startup_data/startup_funding2020.csv')
df21 = pd.read_csv('startup_data/startup_funding2021.csv')

df18.columns

Index(['Company Name', 'Industry', 'Round/Series', 'Amount', 'Location',
      'About Company'],
      dtype='object')

df18.columns=df18.columns.str.lower().str.replace(' ', '_').str.replace('/', '_')
df19.columns=df19.columns.str.lower().str.replace(' ', '_').str.replace('/', '_')
df20.columns=df20.columns.str.lower().str.replace(' ', '_').str.replace('/', '_')
df21.columns=df21.columns.str.lower().str.replace(' ', '_').str.replace('/', '_')

#adding a year column before combining the 4 csv files for better clarity
dfs=[df18,df19,df20,df21]
years=[2018,2019,2020,2021]
for df,year in zip(dfs,years):
    df['year']=year
```

```
#checking the dimensions of the 4 files before combining
for year, df in zip([2018,2019,2020,2021], [df18,df19,df20,df21]):
    print(f"{year}: {df.shape}")

2018: (526, 7)
2019: (89, 10)
2020: (1055, 11)
2021: (1209, 10)

#checking for column names
pd.DataFrame({
    '2018': pd.Series(df18.columns),
    '2019': pd.Series(df19.columns),
    '2020': pd.Series(df20.columns),
    '2021': pd.Series(df21.columns)
})
```

	2018	2019	2020	2021
0	company_name	company_brand	company_brand	company_brand
1	industry	founded	founded	founded
2	round_series	headquarter	headquarter	headquarter
3	amount	sector	sector	sector
4	location	what_it_does	what_it_does	what_it_does
5	about_company	founders	founders	founders
6	year	investor	investor	investor
7	NaN	amount(\$)	amount(\$)	amount(\$)
8	NaN	stage	stage	stage
9	NaN	year	unnamed: _9	year
10	NaN	NaN	year	NaN

```
#comparing if differently names columns represent the same information
print('location vs headquarter')
print(pd.DataFrame({
    '2018':df18['location'],
    '2019':df19['headquarter'],
    '2020':df20['headquarter'],
    '2021':df21['headquarter']
}).head())

print('\nindustry vs sector')
print(pd.DataFrame({
    '2018':df18['industry'],
    '2019':df19['sector'],
    '2020':df20['sector'],
    '2021':df21['sector']
}).head())

print('\nround series vs stage')
print(pd.DataFrame({
    '2018':df18['round_series'],
    '2019':df19['stage'],
    '2020':df20['stage'],
    '2021':df21['stage']
}).head())

print('\namount vs amount($)')
print(pd.DataFrame({
    '2018':df18['amount'],
    '2019':df19['amount($)'],
    '2020':df20['amount($)'],
    '2021':df21['amount($)']
}).head())
```

```
location vs headquarter

      2018      2019      2020      2021
0  Bangalore, Karnataka, India      NaN  Chennai  Bangalore
1    Mumbai, Maharashtra, India    Mumbai  Bangalore    Mumbai
2      Gurgaon, Haryana, India    Mumbai    Pune    Mumbai
3    Noida, Uttar Pradesh, India    Chennai  New Delhi    Mumbai
4  Hyderabad, Andhra Pradesh, India  Telangana    Indore  Gurugram
```

```

industry vs sector
2018
0 Brand Marketing, Event Promotion, Marketing, S... Ecommerce
1 Agriculture, Farming Edtech
2 Credit, Financial Services, Lending, Marketplace Edtech
3 Financial Services, FinTech Interior design
4 E-Commerce Platforms, Retail, SaaS AgriTech

```

```

2020 2021
0 AgriTech AI startup
1 EdTech EdTech
2 Hygiene management EdTech
3 Escrow B2B E-commerce
4 AgriTech FinTech

```

```

round series vs stage
2018 2019 2020 2021
0 Seed NaN NaN Pre-series A
1 Seed Series C Pre-seed NaN
2 Series A Fresh funding Pre-seed Series D
3 Angel Series D NaN Series C
4 Seed NaN NaN Seed

```

```

amount vs amount($)
2018 2019 2020 2021
0 250000 $6,300,000 $200,000 $1,200,000
1 ₹40,000,000 $150,000,000 $100,000 $120,000,000
2 ₹65,000,000 $28,000,000 Undisclosed $30,000,000
3 2000000 $30,000,000 $400,000 $51,000,000
4 - $6,000,000 $340,000 $2,000,000

```

#investigating the extra column in df20

```
df20['unnamed:_9'].isnull().sum()
```

```
np.int64(1053)
```

```
df20[df20['unnamed:_9'].notna()]
```

	company_brand	founded	headquarter	sector	what_it_does	founders	investor	amount(\$)	stage
611	Walrus	2019	Bangalore	Fintech	It provides banking solutions for teens and yo...	Bhagaban Behera, Sriharsha Shetty, Nakul Kelkar	Better Capital	Undisclosed	Pre-Seed
613	goDutch	NaN	Mumbai	Fintech	Group Payments platform	Aniruddh Singh, Riyaz Khan, Sagar Sheth	Matrix India, Y Combinator, Global Founders Ca...	\$1,700,000	Seed Round

```
df20.drop(columns=['unnamed:_9'], inplace=True)
```

Observations

- The differences in number of columns is due to the schema inconsistencies rather than entirely new features.
- Similar attributes are represented using different column names across years.\ eg:\ company_name → company_brand\ industry → sector\ round_series → stage\ amount → amount(\$)\ location → headquarter\ about_company → what_it_does
- The datasets for 2019–2021 contain additional columns (founded, founders, and investor) that are absent in the 2018 dataset.
- The dataset for 2020 had an additional column 'unnamed:_9' which is mostly null values. On further inspection of the two non-null entries, it is found that they are the duplicates of the 'stage' value and hence is dropped

#renaming columns before combining

```
df18.rename(columns={'company_name':'company_brand', 'industry':'sector',
                    'round_series':'stage', 'amount':'amount($)',
                    'location':'headquarter',
                    'about_company':'what_it_does'},inplace=True)
```

#combining the files and extracting only common attributes

```
df=pd.concat([df18,df19,df20,df21],ignore_index=True)
```

```
df=df[['company_brand','sector','stage','amount($)','headquarter','what_it_does','year']]
```

```
df.head()
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year
0	TheCollegeFever	Brand Marketing, Event Promotion, Marketing, S...	Seed	250000	Bangalore, Karnataka, India	TheCollegeFever is a hub for fun, fiesta and f...	2018
1	Happy Cow Dairy	Agriculture, Farming	Seed	₹40,000,000	Mumbai, Maharashtra, India	A startup which aggregates milk from dairy far...	2018
2	MyLoanCare	Credit, Financial Services, Lending, Marketplace	Series A	₹65,000,000	Gurgaon, Haryana, India	Leading Online Loans Marketplace in India	2018
3	PayMe India	Financial Services, FinTech	Angel	2000000	Noida, Uttar Pradesh, India	PayMe India is an innovative FinTech organizat...	2018
4	Eunimart	E-Commerce Platforms, Retail, SaaS	Seed	—	Hyderabad, Andhra Pradesh, India	Eunimart is a one stop solution for merchants ...	2018

```
df.shape
```

```
(2879, 7)
```

```
df.columns
```

```
Index(['company_brand', 'sector', 'stage', 'amount($)', 'headquarter',
      'what_it_does', 'year'],
      dtype='object')
```

Standardizing Company brands

```
df['company_brand']=df['company_brand'].astype(str).str.strip().str.lower().str.replace(r'^a-z0-9 ','',regex=True).str.replace(r'\s+', ' ', regex=True)
```

```
df['company_brand'].value_counts().head(50)
```

	count
company_brand	
byjus	12
bharatpe	10
nykaa	7
zomato	7
mpl	6
oyo	6
zetwerk	6
spinny	6
vedantu	6
trell	6
cred	5
dunzo	5
inframarket	5
furlenco	5
unacademy	5
capital float	5
slice	5

	count
company_brand	
rebel foods	5
teachmint	5
jumbotail	5
licious	5
purplle	4
dealshare	4
servify	4
origo	4
credgenics	4
pristyn care	4
aye finance	4
delhivery	4
pine labs	4
eggoz	4
lido learning	4
swiggy	4
bizongo	4
gomechanic	4
euler motors	4
ninjacart	4
myglamm	4
sarva	4
chingari	4
credr	4
pumpumpum	4
digit insurance	4
yap	4
letstransport	4
satya microcapital	4
true balance	4
mamaearth	4
ola electric	4
instoried	4

dtype: int64

```
df[df['company_brand'].str.contains('byju', na=False)]
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year
130	byjus	EdTech, Education, Higher Education, Secondary...	Private Equity	\$540,000,000	Bangalore, Karnataka, India	BYJU's is an edtech company that is reinventin...	2018
542	byjus	Edtech	NaN	\$540,000,000	NaN	Provides online learning classes	2019
739	byjus	EdTech	NaN	\$200,000,000	Bangalore	BYJU'S is an educational technology company th...	2020
941	byjus	EdTech	NaN	\$500,000,000	Bangalore	An Indian educational technology	2020

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year
						and online tu...	
977	byjus	EdTech	NaN	\$500,000,000	Bangalore	An Indian educational technology and online tu...	2020
1109	byjus	EdTech	NaN	\$122,000,000	Bangalore	Provides online learning classes	2020
1233	byjus	Edtech	NaN	Undisclosed	Bangalore	Provides online learning classes	2020
1550	byju	Edtech	NaN	\$200,000,000	NaN	Provides online learning classes	2020
1650	byju	Edtech	NaN	\$200,000,000	NaN	Provides online learning classes	2020
2308	byjus	EdTech	NaN	\$350000000	Bangalore	BYJU'S is an educational technology company th...	2021
2335	byjus	EdTech	NaN	\$50000000	Bangalore	BYJU'S is an educational technology company th...	2021
2406	byjus	EdTech	Series F	\$460,000,000	Bangalore	BYJU'S is an educational technology company th...	2021
2532	byjus	EdTech	NaN	\$150,000,000	Bangalore	BYJU'S is an educational technology company th...	2021
2621	byjus	EdTech	NaN	\$300000000	Bangalore	BYJU'S is India's largest ed-tech company and ...	2021

```
df['company_brand']=df['company_brand'].replace({'byju':'byjus'})
```

```
df['company_brand'].value_counts().head(50)
```

	count
company_brand	
byjus	14
bharatpe	10
zomato	7
nykaa	7
spinny	6
zetwerk	6
mpl	6
vedantu	6
trell	6
oyo	6
furlenco	5
dunzo	5
jumbotail	5
teachmint	5

	count
company_brand	
inframarket	5
cred	5
slice	5
unacademy	5
rebel foods	5
licious	5
capital float	5
livspace	4
pine labs	4
purpille	4
sarva	4
credr	4
euler motors	4
expertrons	4
mamaearth	4
sharechat	4
servify	4
myglamm	4
yap	4
captain fresh	4
chingari	4
bizongo	4
digit insurance	4
ola electric	4
pitstop	4
origo	4
satya microcapital	4
pristyn care	4
homelane	4
karbon card	4
groww	4
f5	4
lido learning	4
egggoz	4
classplus	4
ninjacart	4

dtype: int64

Insight

- Company names contained inconsistencies due to capitalization and special characters used
- Hence, they were standardized by by converting them to lower case and removing those special characters
- A sanity check was done using byjus to check if the different variations are handled

Standardizing Sectors

```
df['sector']=df['sector'].astype(str).str.split(',').str[0]
df['sector'].value_counts().head(25)
```

	count
sector	
FinTech	175
EdTech	148
Financial Services	88
Fintech	85
Edtech	74
E-commerce	73
Automotive	54
AgriTech	43
Food & Beverages	39
Logistics	37
Healthcare	36
Information Technology & Services	36
Gaming	36
Healthtech	33
HealthCare	31
—	30
SaaS	30
Consumer Goods	29
SaaS startup	28
Finance	27
HealthTech	27
Media	27
Retail	26
Health Care	26
E-learning	25

dtype: int64

#Standardising the top common sectors

```
sector_map = {
    'FinTech': 'Fintech',
    'EdTech': 'Edtech',
    'E-learning': 'Edtech',
    'HealthCare': 'Healthcare',
    'Health Care': 'Healthcare',
    'HealthTech': 'Healthtech',
    'SaaS startup': 'SaaS',
    'Saas': 'SaaS',
    'Ai Startup': 'AI Startup',
    '-': np.nan #replacing - with nulls
}

df['sector'] = df['sector'].str.strip().replace(sector_map)

df['stage'].unique()
```

```
array(['Seed', 'Series A', 'Angel', 'Series B', 'Pre-Seed',
      'Private Equity', 'Venture - Series Unknown', 'Grant',
      'Debt Financing', 'Post-IPO Debt', 'Series H', 'Series C',
      'Series E', 'Corporate Round', 'Undisclosed',
      'https://docs.google.com/spreadsheets/d/1x9ziNeaz6auNChIHnMI8U6kS7knTr3byy_YB',
      'Series D', 'Secondary Market', 'Post-IPO Equity',
      'Non-equity Assistance', 'Funding Round', nan, 'Fresh funding',
      'Pre series A', 'Series G', 'Post series A', 'Seed funding',
      'Seed fund', 'Series F', 'Series B+', 'Seed round', 'Pre-series A',
      'Pre-seed', 'Pre-series', 'Debt', 'Pre-series C', 'Pre-series B',
      'Bridge', 'Series B2', 'Pre- series A', 'Edge', 'Pre-Series B',
      'Seed A', 'Series A-1', 'Seed Funding', 'Pre-seed Round',
      'Seed Round & Series A', 'Pre Series A', 'Pre seed Round',
      'Angel Round', 'Pre series A1', 'Series E2', 'Seed Round',
      'Bridge Round', 'Pre seed round', 'Pre series B', 'Pre series C',
```

```
'Seed Investment', 'Series D1', 'Mid series', 'Series C, D',
'$1200000', 'Seed+', 'Series F2', 'Series A+', 'Series B3', 'PE',
'Series F1', 'Pre-series A1', '$300000', 'Early seed', '$6000000',
'$1000000', 'Seies A', 'Series A2', 'Series I'], dtype=object)
```

Insight

- The sector column contained multivalued cells separated by commas
- Only the first value was retained based on the assumption that they represent the primary sector, while the subsequent values represent additional descriptions
- The variations (like FinTech and Fintech) was also handled

Standardizing Funding Stages

```
df['stage'] = (
    df['stage']
    .astype(str)
    .str.lower()
    .str.strip()
    .str.replace('-', ' ', regex=False)
    .str.replace('/', ' ', regex=False)
    .str.replace(r'[^a-z0-9]', '', regex=True)
    .str.replace(r'\s+', ' ', regex=True)
)
df['stage'].unique()

array(['seed', 'series a', 'angel', 'series b', 'pre seed',
       'private equity', 'venture series unknown', 'grant',
       'debt financing', 'post ipo debt', 'series h', 'series c',
       'series e', 'corporate round', 'undisclosed',
       'https docsgooglecom spreadsheets d
1x9zineaz6aunchihmi8u6ks7kntr3byyybgfqaoua editgid1861303593',
       'series d', 'secondary market', 'post ipo equity',
       'non equity assistance', 'funding round', 'nan', 'fresh funding',
       'pre series a', 'series g', 'post series a', 'seed funding',
       'seed fund', 'series f', 'seed round', 'pre series', 'debt',
       'pre series c', 'pre series b', 'bridge', 'series b2', 'edge',
       'seed a', 'series a 1', 'pre seed round', 'seed round series a',
       'angel round', 'pre series a1', 'series e2', 'bridge round',
       'seed investment', 'series d1', 'mid series', 'series c d',
       '1200000', 'series f2', 'series b3', 'pe', 'series f1', '300000',
       'early seed', '6000000', '1000000', 'seies a', 'series a2',
       'series i'], dtype=object)

stage_map = {
    # Seed
    'seed funding': 'seed',
    'seed fund': 'seed',
    'seed round': 'seed',
    'seed investment': 'seed',
    'early seed': 'seed',
    'seed a': 'seed',

    # Angel
    'angel round': 'angel',

    # Pre-seed
    'pre seed round': 'pre seed',

    # Pre-series A
    'pre series a1': 'pre series a',

    # Series variants
    'series a 1': 'series a',
    'series a2': 'series a',
    'series b2': 'series b',
    'series b3': 'series b',
    'series c d': 'series c',
    'series d1': 'series d',
    'series e2': 'series e',
    'series f1': 'series f',
    'series f2': 'series f',

    # Other
    'bridge round': 'bridge',
```

```

'debt': 'debt financing',
'pe': 'private equity',
'seies a': 'series a',
'post series a': 'series a',
'funding round': 'undisclosed',
'fresh funding': 'undisclosed',
'seed round series a': 'series a',
'mid series': 'venture series unknown',
'edge': np.nan,

# String nan
'nan': np.nan
}

df['stage'] = df['stage'].replace(stage_map)

junk = [
    '1200000',
    '300000',
    '1000000',
    '6000000',
    'https docs.google.com spreadsheets d
    1x9zineaz6aunchihnmi8u6ks7kntr3byyybgfqaoua editgid1861303593'
]

df.loc[df['stage'].isin(junk), 'stage'] = np.nan

df['stage'].value_counts()

```

	count
stage	
seed	695
series a	311
pre series a	291
series b	138
series c	115
pre seed	73
debt financing	58
series d	52
angel	41
venture series unknown	38
series e	31
pre series b	18
series f	13
private equity	11
bridge	9
corporate round	8
undisclosed	4
grant	4
series h	4
pre series c	3
pre series	3
secondary market	3
post ipo equity	3
series g	3
post ipo debt	2
non equity assistance	1
series i	1

dtype: int64

Insight

- The Stage column had a huge number of inconsistencies in formatting, naming convention and spelling differences
- Obvious invalid values like URLs and numeric values were assigned null values

Standardizing Funding Amount

```
# first clean symbols
df['amount($)'] = (
    df['amount($)']
    .astype(str)
    .str.replace('₹', '', regex=False)
    .str.replace('$', '', regex=False)
    .str.replace(',', '', regex=False)
    .replace(['Undisclosed', '-', 'nan'], np.nan)
)

# numeric conversion
df['amount($)'] = pd.to_numeric(df['amount($)'], errors='coerce')

# convert only 2018 rows from INR to USD
df.loc[df['year'] == 2018, 'amount($)'] /= 68.41
df['amount($)'] = df['amount($)'].round()

df.head()
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year
0	thecollegefever	Brand Marketing	seed	3654.0	Bangalore, Karnataka, India	TheCollegeFever is a hub for fun, fiesta and f...	2018
1	happy cow dairy	Agriculture	seed	584710.0	Mumbai, Maharashtra, India	A startup which aggregates milk from dairy far...	2018
2	myloancare	Credit	series a	950153.0	Gurgaon, Haryana, India	Leading Online Loans Marketplace in India	2018
3	payme india	Financial Services	angel	29235.0	Noida, Uttar Pradesh, India	PayMe India is an innovative FinTech organizat...	2018
4	eunimart	E-Commerce Platforms	seed	NaN	Hyderabad, Andhra Pradesh, India	Eunimart is a one stop solution for merchants ...	2018

Insight

- The amount column was recorded with different currencies (using INR in 2018 and USD in 2019-2021)
- The currency symbols and the formatting was removed
- The 2018 amounts were converted to USD with an approximate exchange rate of ₹68.41 per USD

Standardizing Cities

```
df['city'] = df['headquarter'].str.strip().str.split(',').str[0]

df['city'].unique()

array(['Bangalore', 'Mumbai', 'Gurgaon', 'Noida', 'Hyderabad',
       'Bengaluru', 'Kalkaji', 'Delhi', 'India', 'Hubli', 'New Delhi',
       'Chennai', 'Mohali', 'Kolkata', 'Pune', 'Jodhpur', 'Kanpur',
       'Ahmedabad', 'Azadpur', 'Haryana', 'Cochin', 'Faridabad', 'Jaipur',
       'Kota', 'Anand', 'Bangalore City', 'Belgaum', 'Thane', 'Margão',
       'Indore', 'Alwar', 'Kannur', 'Trivandrum', 'Ernakulam',
       'Kormangala', 'Uttar Pradesh', 'Andheri', 'Mylapore', 'Ghaziabad',
       'Kochi', 'Powai', 'Guntur', 'Kalpakkam', 'Bhopal', 'Coimbatore',
       'Worli', 'Alleppey', 'Chandigarh', 'Guindy', 'Lucknow', nan,
       'Telangana', 'Gurugram', 'Surat', 'Uttar pradesh', 'Rajasthan',
```

```
'Tirunelveli', 'Singapore', 'Gujarat', 'Kerala', 'Frisco',
'California', 'Dhingsara', 'New York', 'Patna', 'San Francisco',
'San Ramon', 'Paris', 'Plano', 'Sydney', 'San Francisco Bay Area',
'Bangaldesh', 'London', 'Milano', 'Palmwoods', 'France',
'Samastipur', 'Irvine', 'Tumkur', 'Newcastle Upon Tyne',
'Shanghai', 'Jiaxing', 'Rajasthan', 'Ludhiana', 'Dehradun',
'San Franciscas', 'Tangerang', 'Berlin', 'Seattle', 'Riyadh',
'Seoul', 'Bangkok', 'Warangal', 'Hyderebad', 'Odisha', 'Bihar',
'Goa', 'Tamil Nadu', 'Banglore', 'Ahmadabad', 'Small Towns',
'Rajsamand', 'Ranchi', 'Computer Games', 'Vadodara',
'Food & Beverages', 'Pharmaceuticals\t#REF!', 'Gurugram\t#REF!',
'Nagpur', 'West Bengal', 'Samsitpur', 'Silvassa',
'Thiruvananthapuram', 'Roorkee', 'Ambernath', 'Panchkula',
'Mangalore', 'Telugana', 'Bhubaneswar', 'Kottayam', 'Beijing',
'Panaji', 'Satara', 'Orissia', 'Santra', 'Mountain View',
'Jharkhand', 'Bhilwara', 'Guwahati', 'Online Media\t#REF!',
'Information Technology & Services', 'The Nilgiris', 'Gandhinagar'],
dtype=object)
```

```
df[df['city']=='Online Media\t#REF!']
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
2770	sochcast	Sochcast is an Audio experiences company that ...	NaN	NaN	Online Media\t#REF!	CA Harvinderjit Singh Bhatia, Garima Surana, A...	2021	Online Media\t#REF!

```
df[df['city']=='Computer Games']
#headquater contains sector values
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
1768	fanplay	Computer Games	NaN	NaN	Computer Games	A real money game app specializing in trivia g...	2021	Computer Games
1781	fanplay	Computer Games	NaN	NaN	Computer Games	A real money game app specializing in trivia g...	2021	Computer Games

```
df[df['city']=='Information Technology & Services']
#swapped sector and city/headquarter values
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
2846	peak	Manchester	series c	75000000.0	Information Technology & Services	Peak helps the world's smartest companies put ...	2021	Information Technology & Services

```
df['city'].unique()
```

```
array(['Bangalore', 'Mumbai', 'Gurgaon', 'Noida', 'Hyderabad',
'Bengaluru', 'Kalkaji', 'Delhi', 'India', 'Hubli', 'New Delhi',
'Chennai', 'Mohali', 'Kolkata', 'Pune', 'Jodhpur', 'Kanpur',
'Ahmedabad', 'Azadpur', 'Haryana', 'Cochin', 'Faridabad', 'Jaipur',
'Kota', 'Anand', 'Bangalore City', 'Belgaum', 'Thane', 'Margão',
'Indore', 'Alwar', 'Kannur', 'Trivandrum', 'Ernakulam',
'Kormangala', 'Uttar Pradesh', 'Andheri', 'Mylapore', 'Ghaziabad',
'Kochi', 'Powai', 'Guntur', 'Kalpakkam', 'Bhopal', 'Coimbatore',
'Worli', 'Alleppey', 'Chandigarh', 'Guindy', 'Lucknow', nan,
'Telangana', 'Gurugram', 'Surat', 'Uttar pradesh', 'Rajasthan',
'Tirunelveli', 'Singapore', 'Gujarat', 'Kerala', 'Frisco',
'California', 'Dhingsara', 'New York', 'Patna', 'San Francisco',
'San Ramon', 'Paris', 'Plano', 'Sydney', 'San Francisco Bay Area',
'Bangaldesh', 'London', 'Milano', 'Palmwoods', 'France',
'Samastipur', 'Irvine', 'Tumkur', 'Newcastle Upon Tyne',
'Shanghai', 'Jiaxing', 'Rajasthan', 'Ludhiana', 'Dehradun',
'San Franciscas', 'Tangerang', 'Berlin', 'Seattle', 'Riyadh',
'Seoul', 'Bangkok', 'Warangal', 'Hyderebad', 'Odisha', 'Bihar',
```

```

        'Goa', 'Tamil Nadu', 'Banglore', 'Ahmadabad', 'Small Towns',
        'Rajsamand', 'Ranchi', 'Computer Games', 'Vadodara',
        'Food & Beverages', 'Pharmaceuticals\t#REF!', 'Gurugram\t#REF!',
        'Nagpur', 'West Bengal', 'Samsitpur', 'Silvassa',
        'Thiruvananthapuram', 'Roorkee', 'Ambernath', 'Panchkula',
        'Mangalore', 'Telugana', 'Bhubaneswar', 'Kottayam', 'Beijing',
        'Panaji', 'Satara', 'Orissia', 'Santra', 'Mountain View',
        'Jharkhand', 'Bhilwara', 'Guwahati', 'Online Media\t#REF!',
        'Information Technology & Services', 'The Nilgiris', 'Gandhinagar'],
dtype=object)

random=['Computer Games','Food &
        Beverages','Pharmaceuticals\t#REF!','Gurugram\t#REF!','Online
        Media\t#REF!','Information Technology & Services']
print('Number of rows with random
        cities:',df[df['city'].isin(random)].shape[0])
print('Percent of corrupt cities:',(9/2879)*100)

#dropping the random city values
df = df[~df['city'].isin(random)]

Number of rows with random cities: 9
Percent of corrupt cities: 0.3126085446335533

df['city'].unique()

array(['Bangalore', 'Mumbai', 'Gurgaon', 'Noida', 'Hyderabad',
        'Bengaluru', 'Kalkaji', 'Delhi', 'India', 'Hubli', 'New Delhi',
        'Chennai', 'Mohali', 'Kolkata', 'Pune', 'Jodhpur', 'Kanpur',
        'Ahmedabad', 'Azadpur', 'Haryana', 'Cochin', 'Faridabad', 'Jaipur',
        'Kota', 'Anand', 'Bangalore City', 'Belgaum', 'Thane', 'Margão',
        'Indore', 'Alwar', 'Kannur', 'Trivandrum', 'Ernakulam',
        'Kormangala', 'Uttar Pradesh', 'Andheri', 'Mylapore', 'Ghaziabad',
        'Kochi', 'Powai', 'Guntur', 'Kalpakkam', 'Bhopal', 'Coimbatore',
        'Worli', 'Alleppey', 'Chandigarh', 'Guindy', 'Lucknow', nan,
        'Telangana', 'Gurugram', 'Surat', 'Uttar pradesh', 'Rajasthan',
        'Tirunelveli', 'Singapore', 'Gujarat', 'Kerala', 'Frisco',
        'California', 'Dhingsara', 'New York', 'Patna', 'San Francisco',
        'San Ramon', 'Paris', 'Plano', 'Sydney', 'San Francisco Bay Area',
        'Bangladesh', 'London', 'Milano', 'Palmwoods', 'France',
        'Samastipur', 'Irvine', 'Tumkur', 'Newcastle Upon Tyne',
        'Shanghai', 'Jiaxing', 'Rajasthan', 'Ludhiana', 'Dehradun',
        'San Franciscas', 'Tangerang', 'Berlin', 'Seattle', 'Riyadh',
        'Seoul', 'Bangkok', 'Warangal', 'Hyderebad', 'Odisha', 'Bihar',
        'Goa', 'Tamil Nadu', 'Banglore', 'Ahmadabad', 'Small Towns',
        'Rajsamand', 'Ranchi', 'Vadodara', 'Nagpur', 'West Bengal',
        'Samsitpur', 'Silvassa', 'Thiruvananthapuram', 'Roorkee',
        'Ambernath', 'Panchkula', 'Mangalore', 'Telugana', 'Bhubaneswar',
        'Kottayam', 'Beijing', 'Panaji', 'Satara', 'Orissia', 'Santra',
        'Mountain View', 'Jharkhand', 'Bhilwara', 'Guwahati',
        'The Nilgiris', 'Gandhinagar'], dtype=object)

invalid = [
        'India', 'Haryana', 'Kerala', 'Odisha', 'Rajasthan', 'Gujarat',
        'Tamil Nadu', 'West Bengal', 'Jharkhand', 'Bihar', 'Goa', 'nan'
]

print('No. of rows with obvious non-cities:
        ',df[df['city'].isin(invalid)].shape[0])
print((165/2890)*100)
df=df[~df['city'].isin(invalid)]

No. of rows with obvious non-cities: 51
5.709342560553633

df['city'].unique()

array(['Bangalore', 'Mumbai', 'Gurgaon', 'Noida', 'Hyderabad',
        'Bengaluru', 'Kalkaji', 'Delhi', 'Hubli', 'New Delhi', 'Chennai',
        'Mohali', 'Kolkata', 'Pune', 'Jodhpur', 'Kanpur', 'Ahmedabad',
        'Azadpur', 'Cochin', 'Faridabad', 'Jaipur', 'Kota', 'Anand',
        'Bangalore City', 'Belgaum', 'Thane', 'Margão', 'Indore', 'Alwar',
        'Kannur', 'Trivandrum', 'Ernakulam', 'Kormangala', 'Uttar Pradesh',
        'Andheri', 'Mylapore', 'Ghaziabad', 'Kochi', 'Powai', 'Guntur',
        'Kalpakkam', 'Bhopal', 'Coimbatore', 'Worli', 'Alleppey',
        'Chandigarh', 'Guindy', 'Lucknow', nan, 'Telangana', 'Gurugram',
        'Surat', 'Uttar pradesh', 'Tirunelveli', 'Singapore', 'Frisco',
        'California', 'Dhingsara', 'New York', 'Patna', 'San Francisco',
        'San Ramon', 'Paris', 'Plano', 'Sydney', 'San Francisco Bay Area',
        'Bangladesh', 'London', 'Milano', 'Palmwoods', 'France',

```

```

'Samastipur', 'Irvine', 'Tumkur', 'Newcastle Upon Tyne',
'Shanghai', 'Jiaxing', 'Rajasthan', 'Ludhiana', 'Dehradun',
'San Francisco', 'Tangerang', 'Berlin', 'Seattle', 'Riyadh',
'Seoul', 'Bangkok', 'Warangal', 'Hyderebad', 'Banglore',
'Ahmadabad', 'Small Towns', 'Rajsamand', 'Ranchi', 'Vadodara',
'Nagpur', 'Samsitpur', 'Silvassa', 'Thiruvananthapuram', 'Roorkee',
'Ambernath', 'Panchkula', 'Mangalore', 'Telugana', 'Bhubaneswar',
'Kottayam', 'Beijing', 'Panaji', 'Satara', 'Orissia', 'Santra',
'Mountain View', 'Bhilwara', 'Guwahati', 'The Nilgiris',
'Gandhinagar'], dtype=object)

city_stand = {
'Bangalore':'Bengaluru',
'Banglore':'Bengaluru',
'Bangalore City':'Bengaluru',
'Hyderebad':'Hyderabad',
'Gurgaon':'Gurugram',
'Ahmadabad':'Ahmedabad',
'Rajasthan':'Rajasthan',
'Telugana':'Telangana',
'Orissia':'Odisha',
'Cochin':'Kochi',
'Trivandrum':'Thiruvananthapuram',
'Samsitpur':'Samastipur',
'San Francisco':'San Francisco',
'New Delhi':'Delhi'
}

df['city'] = df['city'].replace(city_stand)

```

Insight

- The headquarter column contained value ranging from just cities to full addresses including city,state and country
- Only the cities were retained in a new column 'city'
- There were a few invalid records and some states and since they accounted only a small percentage of values, they were dropped with negligible difference

```
df.head()
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
0	thecollegefever	Brand Marketing	seed	3654.0	Bangalore, Karnataka, India	TheCollegeFever is a hub for fun, fiesta and f...	2018	Bengaluru
1	happy cow dairy	Agriculture	seed	584710.0	Mumbai, Maharashtra, India	A startup which aggregates milk from dairy far...	2018	Mumbai
2	myloancare	Credit	series a	950153.0	Gurgaon, Haryana, India	Leading Online Loans Marketplace in India	2018	Gurugram
3	payme india	Financial Services	angel	29235.0	Noida, Uttar Pradesh, India	PayMe India is an innovative FinTech organizat...	2018	Noida
4	eunimart	E-Commerce Platforms	seed	NaN	Hyderabad, Andhra Pradesh, India	Eunimart is a one stop solution for merchants ...	2018	Hyderabad

Data Exploration

```
df.shape
```

```
(2819, 8)
```

```
df.head()
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
0	thecollegefever	Brand Marketing	seed	3654.0	Bangalore, Karnataka,	TheCollegeFever is a hub for fun,	2018	Bengaluru

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
					India	fiesta and f...		
1	happy cow dairy	Agriculture	seed	584710.0	Mumbai, Maharashtra, India	A startup which aggregates milk from dairy far...	2018	Mumbai
2	myloancare	Credit	series a	950153.0	Gurgaon, Haryana, India	Leading Online Loans Marketplace in India	2018	Gurugram
3	payme india	Financial Services	angel	29235.0	Noida, Uttar Pradesh, India	PayMe India is an innovative FinTech organizat...	2018	Noida
4	eunimart	E-Commerce Platforms	seed	NaN	Hyderabad, Andhra Pradesh, India	Eunimart is a one stop solution for merchants ...	2018	Hyderabad

df.info()

```
<class 'pandas.core.frame.DataFrame'>
Index: 2819 entries, 0 to 2878
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   company_brand    2819 non-null   object
1   sector           2789 non-null   object
2   stage            1895 non-null   object
3   amount($)        2277 non-null   float64
4   headquarter      2705 non-null   object
5   what_it_does     2819 non-null   object
6   year             2819 non-null   int64
7   city             2705 non-null   object
dtypes: float64(1), int64(1), object(6)
memory usage: 198.2+ KB
```

df.describe()

	amount(\$)	year
count	2.277000e+03	2819.000000
mean	1.203706e+08	2020.028379
std	3.469238e+09	1.085051
min	1.000000e+01	2018.000000
25%	5.430000e+05	2020.000000
50%	2.411928e+06	2020.000000
75%	1.000000e+07	2021.000000
max	1.500000e+11	2021.000000

df[df['sector'].isnull()].head()

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
58	missmalini entertainment	NaN	seed	1520246.0	Mumbai, Maharashtra, India	MissMalini Entertainment is a multi-platform n...	2018	Mumbai
105	jagaran microfin	NaN	debt financing	8039760.0	Kolkata, West Bengal, India	Jagaran Microfin is a Microfinance institution...	2018	Kolkata
121	fleeca	NaN	seed	NaN	Jaipur, Rajasthan, India	FLEECA is a Tyre Care Provider company.	2018	Jaipur
146	wheelsemi	NaN	series b	204648.0	Pune, Maharashtra, India	WheelsEMI is the brand name of	2018	Pune

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
						NBFC, WheelsEMI...		
153	fric bergen	NaN	venture series unknown	NaN	Alwar, Rajasthan, India	Fric Bergen is a leader in the specialty food ...	2018	Alwar

```
df.isnull().sum()
```

	0
company_brand	0
sector	30
stage	924
amount(\$)	542
headquarter	114
what_it_does	0
year	0
city	114

```
dtype: int64
```

Handling missing values

- Even though the sector column has 30 missing values, the 'what_it_does' column still provides the useful context information
- Missing values in the 'stage' and 'amount(\$)' columns may represent unavailable information rather than data entry errors and hence decided against imputing them

```
df.duplicated().sum()
```

```
np.int64(23)
```

```
df[df.duplicated(keep=False)].sort_values('company_brand').head()
```

	company_brand	sector	stage	amount(\$)	headquarter	what_it_does	year	city
1787	advantage club	HRTech	NaN	1700000.0	Mumbai	Advantage Club is India's largest employee eng...	2021	Mumbai
1774	advantage club	HRTech	NaN	1700000.0	Mumbai	Advantage Club is India's largest employee eng...	2021	Mumbai
1922	asqi advisors	Financial Services	pre series a	1000000.0	Mumbai	Bringing Blockchain technology intro mainstrea...	2021	Mumbai
1908	asqi advisors	Financial Services	pre series a	1000000.0	Mumbai	Bringing Blockchain technology intro mainstrea...	2021	Mumbai
1779	bewakoof	Apparel & Fashion	NaN	8000000.0	Mumbai	Bewakoof is a lifestyle fashion brand that mak...	2021	Mumbai

```
#dropping duplicates
```

```
df.drop_duplicates(inplace=True)
```

```
df.reset_index(drop=True,inplace=True)
```

```
df = df[['company_brand','sector','stage','amount($)','what_it_does','year','city']]
```

```
df.rename(columns={'amount($)':'amount_usd'}, inplace=True)
```

Final Cleaned Dataset Summary

- Total records: 2796
- Features: 8
- Time period: 2018-2021 *Missing values were retained where absence represented genuinely unavailable information rather than data quality issues
- The final dataset was standardized to ensure consistency across years and facilitate further analysis

```
#saving the cleaned dataset for analysis using MySQL
```

```
df.to_csv('startup_cleaned.csv',index=False)
```

```
from google.colab import files
```

```
#files.download('startup_cleaned.csv')
```

```
df
```

	company_brand	sector	stage	amount_usd	what_it_does	year	city
0	thecollegefever	Brand Marketing	seed	3654.0	TheCollegeFever is a hub for fun, fiesta and f...	2018	Bengaluru
1	happy cow dairy	Agriculture	seed	584710.0	A startup which aggregates milk from dairy far..	2018	Mumbai
2	myloancare	Credit	series a	950153.0	Leading Online Loans Marketplace in India	2018	Gurugram
3	payme india	Financial Services	angel	29235.0	PayMe India is an innovative FinTech organizat...	2018	Noida
4	eunimart	E-Commerce Platforms	seed	NaN	Eunimart is a one stop solution for merchants ...	2018	Hyderabad
...	...	...	...	...	...	...	...
2791	gigforce	Staffing & Recruiting	pre series a	3000000.0	A gig/on-demand staffing company.	2021	Gurugram
2792	vahdam	Food & Beverages	series d	2000000.0	VAHDAM is among the world's first vertically i...	2021	Delhi
2793	leap finance	Financial Services	series c	5500000.0	International education loans for high potenti...	2021	Bengaluru
2794	collegedekho	Edtech	series b	2600000.0	Collegedekho.com is Student's Partner, Friend ...	2021	Gurugram
2795	werize	Financial Services	series a	8000000.0	India's first socially distributed full stack ...	2021	Bengaluru

2796 rows × 7 columns